Design Pattern and Principal

Exercise 1: Implementing the Singleton Pattern

```java
public class SingletonTest {

    Run | Debug
    public static void main(String[] args) {

        // Get Logger instances
        Logger logger1 = Logger.getInstance();
        Logger logger2 = Logger.getInstance();

        // Use logger
        logger1.log(message: "Application started.");
        logger2.log(message: "Processing data.");

        // Verify both references point to the same object
        System.out.println("logger1 hashCode: " + logger1.hashCode());
        System.out.println("logger2 hashCode: " + logger2.hashCode());

        if (logger1 == logger2) {
            System.out.println(x: "Only one Logger instance exists.");
        } else {
            System.out.println(x: "Multiple Logger instances exist.");
        }
    }
}
```

```java
public class Logger {

    // Private static instance of Logger
    private static Logger instance;

    // Private constructor to prevent object creation from outside
    private Logger() {
        System.out.println(x: "Logger instance created.");
    }

    // Public static method to provide access to the instance
    public static Logger getInstance() {
        if (instance == null) {
            instance = new Logger();
        }
        return instance;
    }

    // Sample logging method
    public void log(String message) {
        System.out.println("LOG: " + message);
    }
}
```

Exercise 2: Implementing the Factory Method Pattern

```
c:\Users\Lohitha\OneDrive\Desktop\Digital_Nurture_Cts\Week_1\Design_and_principal>java SingletonTest
Logger instance created.
LOG: Application started.
LOG: Processing data.
logger1 hashCode: 1933863327
logger2 hashCode: 1933863327
Only one Logger instance exists.
```

```java
package FactoryMethodPatternExample;

public interface Document {
    void open();
}
```

```java
package FactoryMethodPatternExample;

public class PdfDocument implements Document {

    @Override
    public void open() {
        System.out.println(x: "PDF Document Opened");
    }
}
```

```java
package FactoryMethodPatternExample;

public class ExcelDocument implements Document {

    @Override
    public void open() {
        System.out.println(x: "Excel Document Opened");
    }
}
```

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```java
package FactoryMethodPatternExample;

public class WordDocument implements Document {

    @Override
    public void open() {
        System.out.println(x: "Word Document Opened");
    }
}
```

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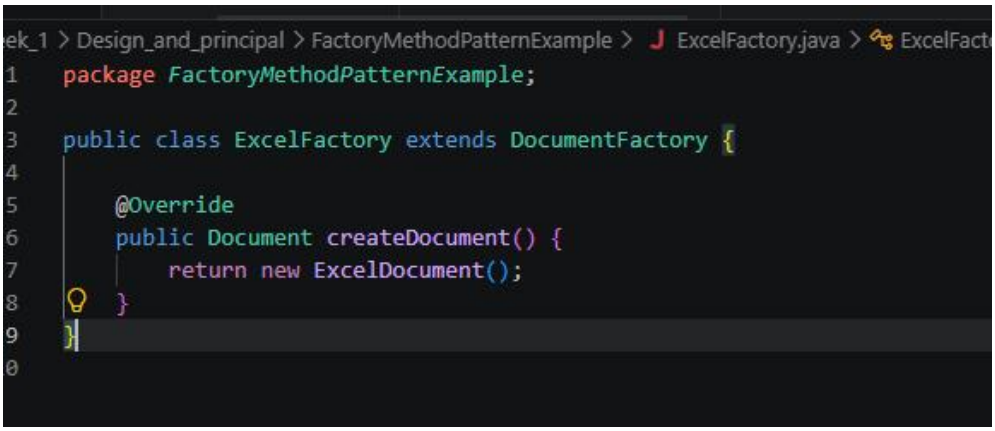
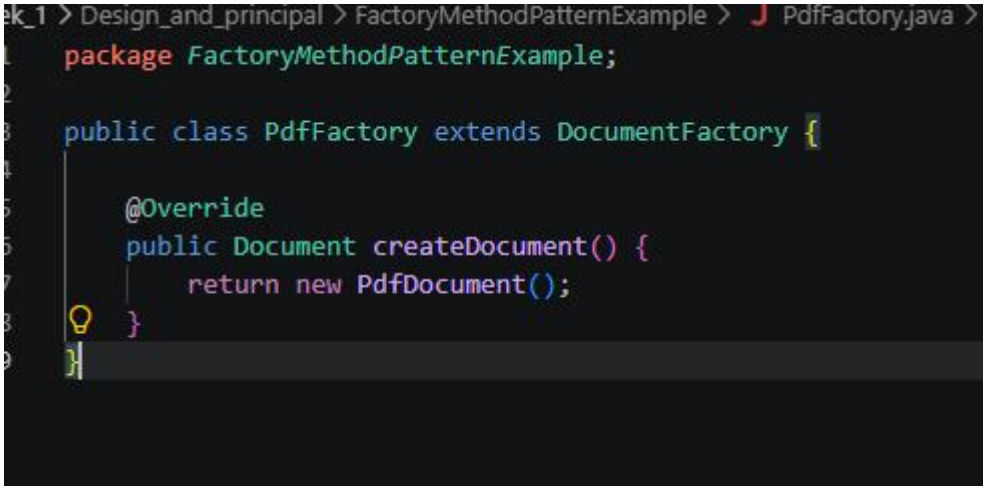
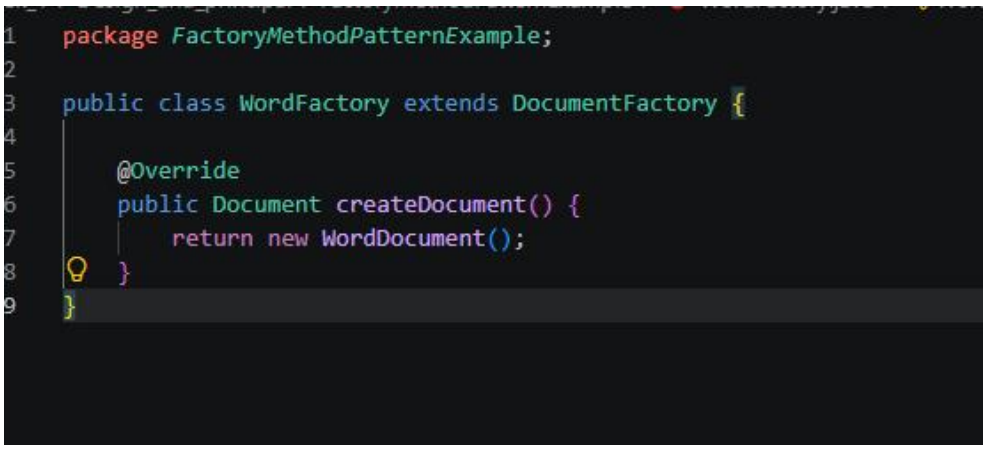
```java
package FactoryMethodPatternExample;

public abstract class DocumentFactory {

    public abstract Document createDocument();
}
```


ek_1 > Design_and_principal > FactoryMethodPatternExample > ExcelFactory.java > ExcelFact
package FactoryMethodPatternExample;
public class ExcelFactory extends DocumentFactory {
@Override
public Document createDocument() {
return new ExcelDocument();
}
}


k_1 > Design_and_principal > FactoryMethodPatternExample > PdfFactory.java >
package FactoryMethodPatternExample;
public class PdfFactory extends DocumentFactory {
@Override
public Document createDocument() {
return new PdfDocument();
}
}


package FactoryMethodPatternExample;
public class WordFactory extends DocumentFactory {
@Override
public Document createDocument() {
return new WordDocument();
}
}

```java
package FactoryMethodPatternExample;

public class FactoryTest {

    Run | Debug
    public static void main(String[] args) {

        DocumentFactory wordFactory = new WordFactory();
        Document wordDoc = wordFactory.createDocument();
        wordDoc.open();

        DocumentFactory pdfFactory = new PdfFactory();
        Document pdfDoc = pdfFactory.createDocument();
        pdfDoc.open();

        DocumentFactory excelFactory = new ExcelFactory();
        Document excelDoc = excelFactory.createDocument();
        excelDoc.open();
    }
}
```

```
c:\Users\Lohitha\OneDrive\Desktop\Digital_Nurture_Cts\Week_1\Design_and_principal>java FactoryMethodPatternExample.FactoryTest
Word Document Opened
PDF Document Opened
Excel Document Opened
```